**ZEISS India Hackathon · Beyond Code 2026**

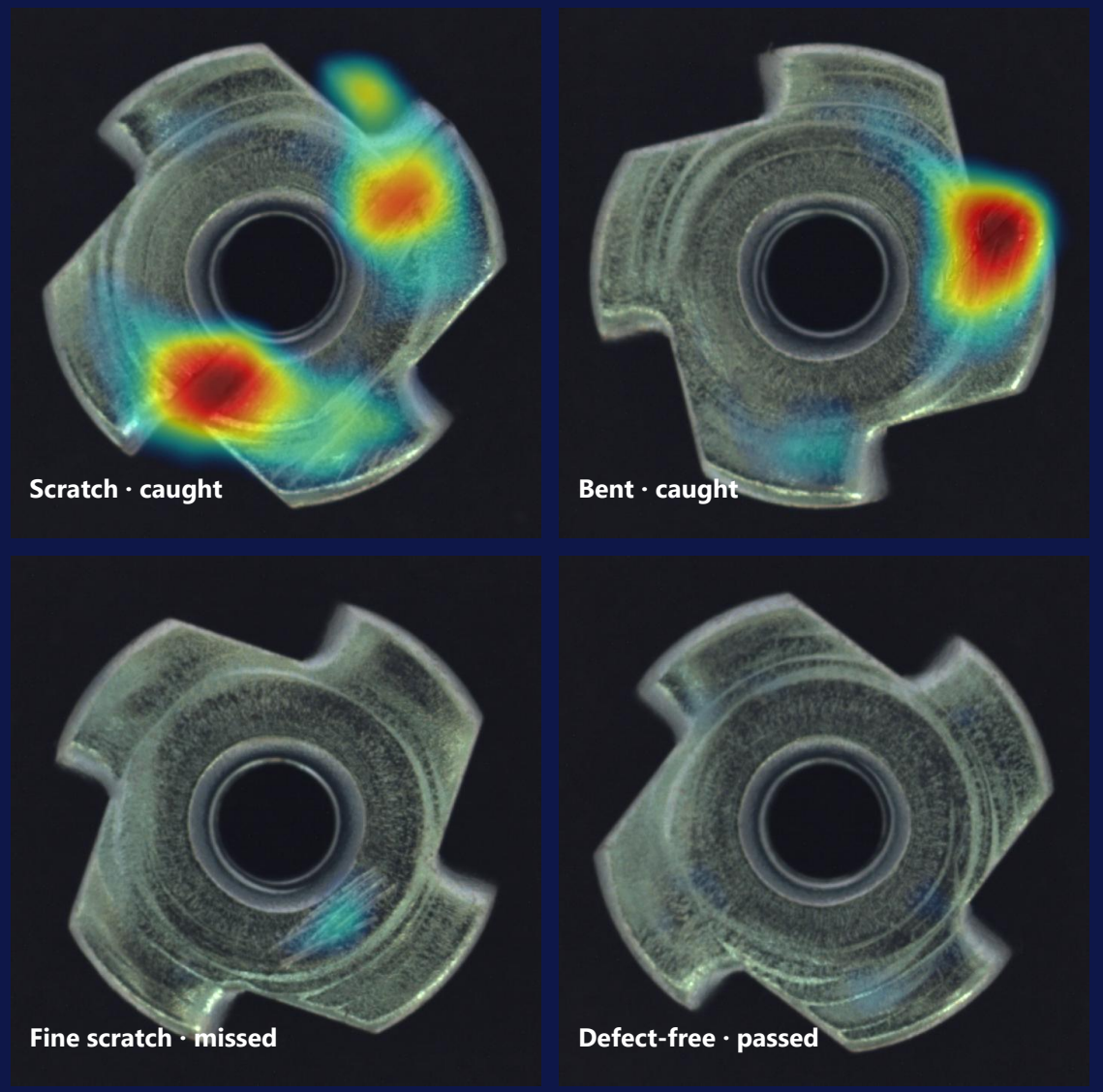
AI/ML for Advanced Frugal Innovation

# NetraQC

## A low-cost inspection concept inspired by ZEISS SurfMax, for India's small factories

**Team Tensor-hive, IIT Kanpur**

A concept proposed by Team Tensor-hive. Not a ZEISS product or a ZEISS-endorsed design.




MVTec AD benchmark images with NetraQC heatmaps

# Learns what normal looks like, then flags surface defects with a heatmap

## What is new

1. **Set up from defect-free images only** — Built
   Pretrained features, matched to a memory of normal parts.
2. **ML picks the leanest design** — Built
   Searched 300 designs; the 131 MB model adds no accuracy.
3. **Uses light from several directions** — Planned
   Sprint test on public multi-light datasets: does it reveal shallow scratches?

**0.985**
image AUROC on the benchmark (1.0 is perfect)

**82 / 93**
defective images caught at our threshold (88%)

**0 / 22**
false alarms on 22 defect-free test images (too few for a rate)

**< 0.1 s**
per image on a laptop CPU; Pi not yet measured

## Our answers to what the organisers ask for

1 **Problem** — Manual inspection in small factories — slide 3
2 **ZEISS product** — SurfMax, and why we chose it — slide 4
3 **Idea** — Design choices, how it works, ML optimiser — slides 5–7
4 **Results** — What it catches and what it misses — slide 8
5 **Platform** — Parts list and software stack — slide 9
6 **Plan** — Sprint steps and expected results — slide 11

Results: our runs on MVTec AD (metal nut), 14–15 Oct 2026. AUROC: how well scores rank defective above defect-free images (1.0 is perfect). Full sources on slide 12.

# Small factories still inspect by eye, and AI inspection doesn't fit them

Surface defects such as scratches and dents are checked by eye in many MSME plants.

## How big

India's MSMEs, Economic Survey 2025-26

**35.4%**

of manufacturing output

**48.6%**

of exports

**7.47 crore**

enterprises

## People miss defects

US study of 82 trained inspectors of precision parts

**Defective parts caught**




**Good parts wrongly rejected**




*It shows that inspection by eye is fallible. It is not a measured rate for Indian factories.*

## Why existing AI rarely fits

What AI inspection needs vs. what a small plant has

| NEEDS | SMALL PLANT HAS |
|---|---|
| **Thousands of labelled defect images** | No defect images |
| **GPU or cloud link** | Dusty floor, patchy internet |
| **Industrial cameras and lighting** | A tight budget |
| **A vision engineer** | No vision engineer |

Typical stations cost roughly $3k–$50k (vendor estimates).

Sources: Economic Survey 2025-26 (MSME shares); See, J.E., Human Factors 57(8), 2015 (82 inspectors: 85% of defective items caught, 35% of good items rejected); Jidoka and iFactory pricing guides (vendor estimates); Dupple AI inspection tools review (2026).

# Inspired by ZEISS SurfMax: a lower-cost route to surface-shape information

The brief asks us to start from an existing ZEISS product. We compared three; NetraQC is our own concept.

## ZEISS products we considered

REJECTED

### VISUSCOUT 100 fundus camera + AI

Retinal AI is crowded and needs patient data to validate.

REJECTED

### X-ray CT + AI fast scanning

ZEISS already offers AI reconstruction (DeepRecon).

CHOSEN

### SurfMax surface inspection

Its use of surface-shape information can be explored with LEDs and tested on public data.

## ZEISS SurfMax vs. NetraQC

| | ZEISS SURFMAX | NETRAQC | STATUS |
|---|---|---|---|
| **Slope measured by** | Deflectometry | Photometric stereo, 8 LEDs | Planned |
| **Channels** | Greyscale, gloss, slope | Greyscale + surface normals | Planned |
| **Learning** | ML classification, severity settings | Pretrained features + defect-free-only memory | Built |
| **Surfaces** | Matte to mirror-like | Matte to semi-gloss; shiny untested | Planned |
| **Set-up** | Inline, robot handling | Bench-top, Raspberry Pi, offline | Planned |
| **Built for** | High-end lines (e.g. aerospace) | Small Indian factories (MSMEs) | Estimate |

**Different methods:** SurfMax uses deflectometry to extract surface information. NetraQC explores photometric stereo with an 8-LED dome as a lower-cost way to estimate surface normals, and will first test on public multi-light datasets whether it improves defect detection.

Sources: ZEISS SurfMax product information and ZEISS Innovation Magazine (2021); Aerospace Testing International (2021); ZEISS press release on DeepRecon (2021); ZEISS VISUSCOUT 100 product page; Rao, B.C., "Advanced frugal innovations", IEEE T&S. We make no claim about SurfMax pricing.

# Each design choice follows from a fact about inspection

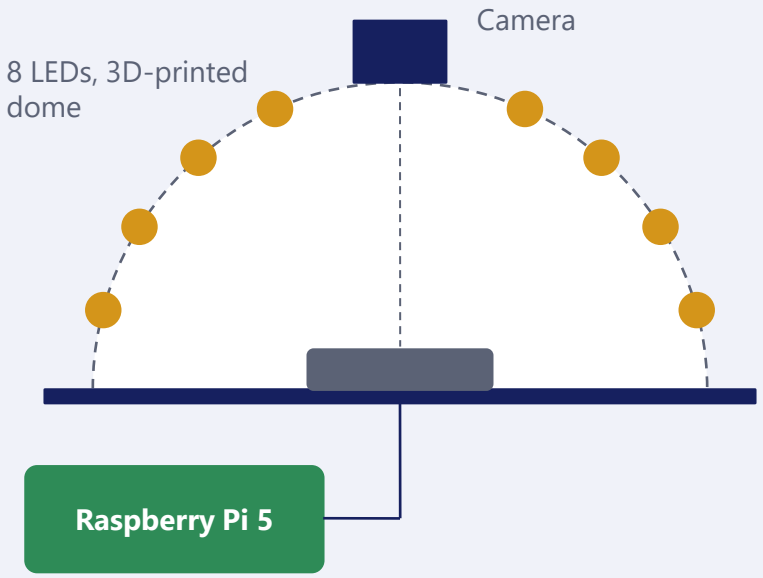
ZEISS defines AFI as frugal through innovative technology, not downgraded technology. Each row makes one part frugal: data, model or hardware.

| FACT | SO WE… | FRUGAL PART | STATUS |
|---|---|---|---|
| Defective parts are rare and vary a lot; defect-free parts are easy to collect. | Build the anomaly memory from defect-free images only; flag anything that deviates from it. | **Data** | **Built** |
| Scratches and dents change surface shape more than colour. | Light the part from several directions and estimate surface normals (photometric stereo). | **Hardware** | **Planned** |
| Spotting a defect needs local texture, not a giant model. | Small pretrained ResNet-18 features plus a compact nearest-match memory, on a CPU. | **Model** | **Built** |
| Lighting and focus drift on a shop floor. | Image a reference card at start-up; compare sharpness and per-LED brightness. | **Hardware** | **Planned** |
| Operators act only on verdicts they can check. | Show an anomaly heatmap with every pass/fail. | **Model** | **Built** |

Sources: ZEISS BeyondCode'26 kick-off, AFI theme brief (Oct 2026); Rao, B.C., "Frugal AI: AI as an Advanced Frugal Innovation!", HAL hal-05727850 (2026). Status: Built = implemented and measured on MVTec AD; Planned = part of the hardware build.

# Set up once from 20 defect-free images, then score every new part

## Proposed station (schematic, not built)




### SET-UP, ONCE PER PART TYPE (≈7 s)

- 20 defect-free images — **Built**
- ResNet-18 patch features (784 per image) — **Built**
- Coreset memory: 1,568 patches, 2.4 MB — **Built**
- Leave-one-out threshold (2.32) — **Built**

### INSPECTION, EVERY PART (<0.1 s ON A LAPTOP)

- Reference-card check; capture under the LEDs — **Planned**
- Surface normals (slope channel) — **Planned**
- Distance to nearest stored patch — **Built**
- Heatmap + PASS/FAIL; log — **Built**

No defective images are used to build the memory or set the threshold (the highest score any training image gets against a memory built from the other 19). Defective test images are used only for evaluation.

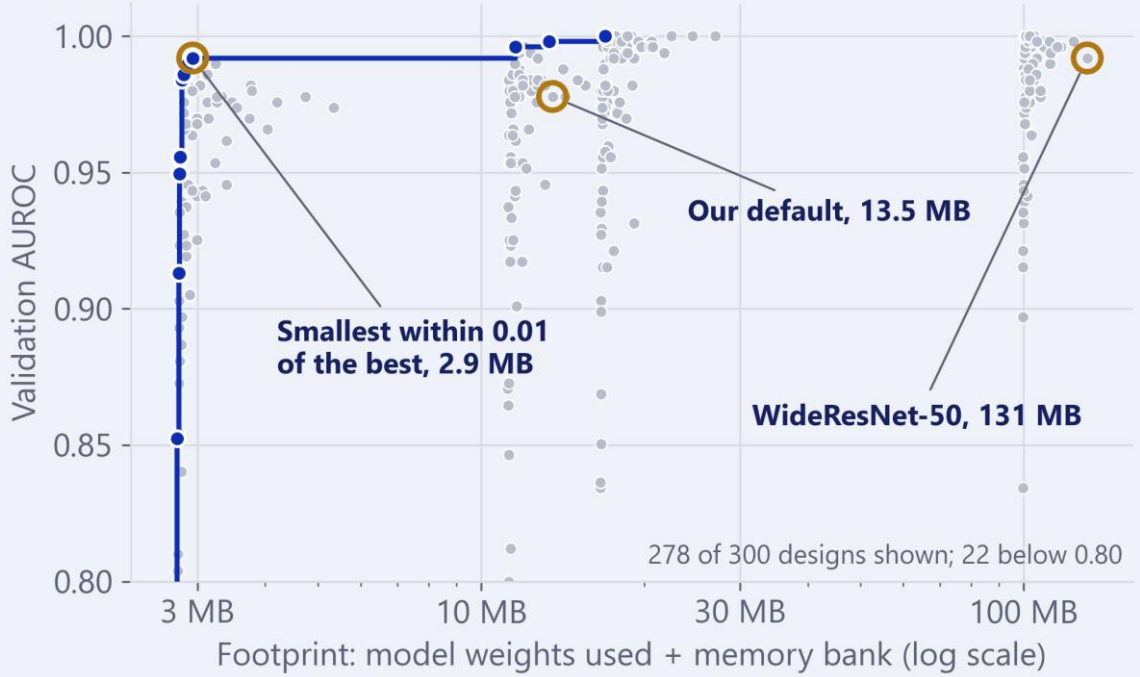
Method adapted from PatchCore: Roth et al., CVPR 2022. Our implementation: netraqc_demo/netraqc_patchcore.py. Timing on a laptop CPU (4 threads), 50–75 ms per image across two runs; Raspberry Pi not yet measured.

# ML design optimiser: the leanest design that keeps accuracy

The AFI brief asks for ML that optimises design parameters for cost or footprint. Built for the detector; next, the lighting.

## 300 detector designs, each measured

Built




Blue: best accuracy reachable at each footprint. Search space: 2 backbones × 5 input sizes × 2 feature depths × 5 memory sizes × 3 training-set sizes.

## Three designs on images the optimiser never saw

| DESIGN | AUROC | MS | FOOTPRINT |
|---|---|---|---|
| WideResNet-50 (paper-style) | 0.992 | 362 | 131 MB |
| **Our default, ResNet-18** | **0.992** | **67** | **13.5 MB** |
| Smallest within 0.01 | 0.977 | 39 | 2.9 MB |

**The 131 MB model buys no accuracy.** Below 3 MB, accuracy drops by 0.015–0.025 (checked on both halves).

**98.8%**

Bayesian optimiser, 40 tries (±0.5)

**96.0%**

Random search, 40 tries (±3.6)

Share of the best trade-off that testing all 300 designs finds

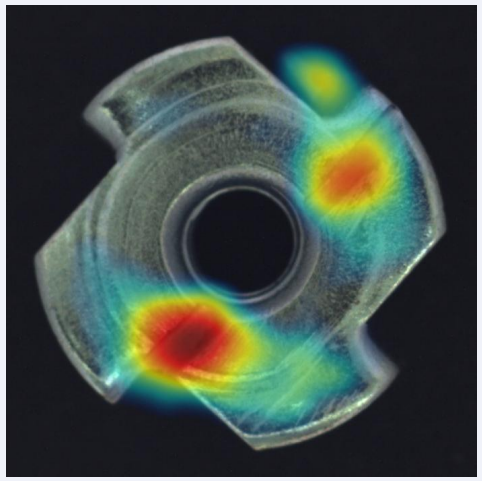
**Next, the lighting, in software.** Public datasets photograph each part under single lights. Light adds linearly, so the optimiser can score every light combination without building hardware.

Our runs, 17 Oct 2026 (netraqc_demo/design_optimizer.py, design_search_eval.py). MVTec AD metal nut test set split 56 validation / 59 held-out; designs chosen on validation. Trade-off = hypervolume over 1−AUROC, log latency, log footprint; 20 seeds. Laptop CPU, 4 threads.

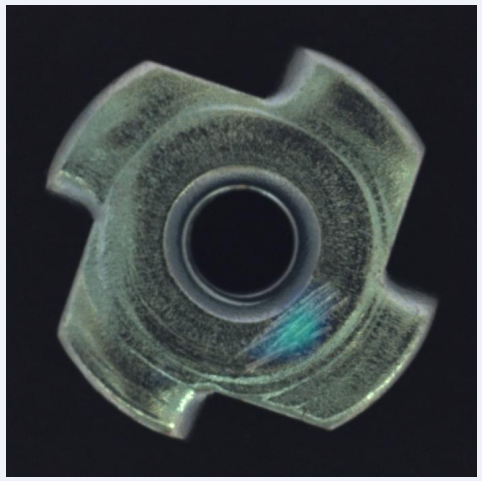
# What the detector catches, what it misses, and why

MVTec AD metal nut: 93 defective and 22 defect-free test images.

## Caught and missed



**Caught.** Deep scratch, scored well above the threshold.



**Missed.** Fine scratches, scored below the threshold. The heatmap points at them, but weakly.

**Fine, shallow scratches like this are what multi-light imaging will test in the sprint.**

**0.985**

Image AUROC: how well image scores separate defective from defect-free images

**0.981**

Pixel AUROC: how well pixel scores separate defect pixels from normal ones

**82 / 93**

defective images caught at the threshold; 0 of 22 defect-free flagged

## Image AUROC by defect type




### The 11 misses

5 scratches, 4 colour stains, 2 bent; 9 scored within 0.17 of the threshold.

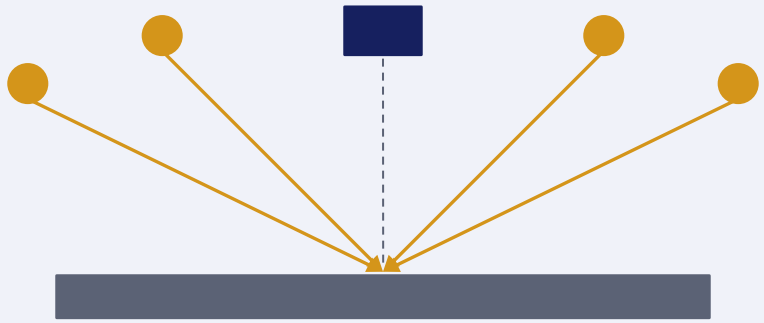
### Limits

One product category of lab images. Not yet run on a Raspberry Pi or on real factory parts.

Dataset: Bergmann et al., MVTec AD, CVPR 2019 (CC BY-NC-SA 4.0, research use only). Our runs, 14–15 Oct 2026; per-image scores in netraqc_demo/results/. AUROC measures how well scores rank defective above defect-free images across all thresholds.

# Platform: open-source software now, a costed ₹12k station as the target

## Estimating surface shape with LEDs



From images under several known light directions, photometric stereo estimates the surface normal at each pixel. SurfMax's deflectometry suits shiny parts; LEDs suit matte and semi-gloss ones.

## Target station (built later)

Indian distributor guides, 2026; not quotes

| | |
|---|---|
| Raspberry Pi 5 (4 GB) | ₹5,500–6,200 |
| Camera Module 3* | ₹2,500–3,000 |
| 8-LED ring + printed dome | ₹900–1,500 |
| Power, microSD, cables | ₹1,700–2,100 |
| **Total** | **₹10,600–12,800** |

*Camera price is our estimate.

## Software stack

**AI**

PyTorch, torchvision, scikit-learn, Optuna

**Imaging**

OpenCV, NumPy

**Edge**

ONNX → OpenVINO INT8, tested on a 4-thread CPU

**App**

FastAPI + SQLite, offline dashboard

**Synthetic data**

BlenderProc (later, to name defect types)

The sprint is software-only: development and benchmarking on a laptop CPU limited to 4 threads, as on a Raspberry Pi 5. The station is built only after the sprint. Shiny or transparent parts may need a diffuser or cross-polarised light.

# No publicly documented low-cost, offline system combines all five

What our desk research found, from public documentation only. NetraQC is shown twice: built today and planned.

| | NO DEFECT IMAGES NEEDED | SURFACE-SLOPE IMAGING | ML DESIGN OPTIMISATION | OFFLINE, UNDER ₹15K | REFERENCE-CARD CHECK |
|---|---|---|---|---|---|
| ZEISS SurfMax (deflectometry) | *Not stated* | **Yes** | *Not stated* | *Not stated* | *Not stated* |
| No-code AI software (e.g. LandingLens) | Partial | No | No | No | No |
| Turnkey vision stations (e.g. Qualitas, Cognex) | Partial | Partial | No | No | Partial |
| Pi + LED research rig (PSBDDS, 2022) | No | **Yes** | No | **Yes** | No |
| **NetraQC today** | **Yes** | No | Partial | No | No |
| **NetraQC as planned** | **Yes** | *Planned* | *Planned* | *Planned* | *Planned* |

**Each piece is published separately. Our contribution is combining them in one low-cost, offline station and testing whether the combination works.**

Sources: ZEISS SurfMax product information; vendor websites; Dupple AI inspection tools review (2026); PSBDDS, PubMed Central PMC9655976 (2022). "Partial" = depends on vendor or configuration. Ratings: public documentation, not product testing. "Not stated" = not established from public documentation. NetraQC today: design search built for software, lighting planned.

# Hackathon Sprint: five measured steps, each with an expected result

Steps run in order and build on the existing detector.

| # | STEP | WHAT WE'LL MEASURE | EXPECTED RESULT |
|---|---|---|---|
| 1 | **Compress the detector for edge use** | ONNX + INT8 (OpenVINO) on a 4-thread CPU: AUROC and time vs. full precision | AUROC within 0.01 of full precision; time per image reported |
| 2 | **Check that light adds linearly** | Eyecandies: sum of the 4 single-light shots vs. the all-lights shot | Small error, so light combinations can be simulated |
| 3 | **Compare single image vs. multi-light** | AUROC on Eyecandies, same parts and split; PS-AD and MSDD if access is granted | Measured change in AUROC and capture time; keep multi-angle only if the gain justifies the time |
| 4 | **Run the ML design optimiser on the lighting** | Light subsets and detector settings together; AUROC, captures per part, cost | Fewest lights within 0.01 AUROC of all lights |
| 5 | **Test on all 15 MVTec AD categories** | Defects caught and false alarms per category | Every category reported, with an image of every miss |

## What would change our plan

**If multi-angle lighting doesn't beat a single image,** the target station keeps one camera shot and uses the LEDs as diffuse light. **If INT8 loses accuracy,** we keep full precision and use the optimiser to pick a smaller input instead.

## Top risks and mitigation

| | |
|---|---|
| **Shiny or transparent parts** | Diffuser or cross-polarised light; test |
| **Misses vs. false alarms** | Adjustable threshold; borderline parts to a person |
| **Benchmark ≠ factory** | 15 categories now; real parts on the built station |

Hackathon Sprint: 1–8 November 2026, software only (slide 9). Step 4 extends the ML design optimiser on slide 7 to lighting. Eyecandies: Bonfiglioli et al., ACCV 2022.

# Sources


- Rao, B.C. (2026). Frugal AI: AI as an Advanced Frugal Innovation! HAL preprint hal-05727850.
- Rao, B.C. Advanced frugal innovations. IEEE Technology and Society (technologyandsociety.org).
- ZEISS. SurfMax surface defect inspection: product information (zeiss.com); ZEISS Innovation Magazine, “Inline defect detection” (2021).
- Aerospace Testing International (2021). Surface defect detection for aerospace parts (ZEISS SurfMax R-Series).
- ZEISS (2021). Press release: ZEISS applies artificial intelligence to 3D X-ray microscope reconstruction (DeepRecon).
- ZEISS. VISUSCOUT 100 handheld fundus camera: product page (zeiss.com).
- Government of India (2026). Economic Survey 2025-26: MSME share of manufacturing output and exports.
- See, J.E. (2015). Visual inspection reliability for precision manufactured parts. Human Factors 57(8), 1427–1442.
- Jidoka Technologies and iFactory (2026). AI visual inspection pricing guides (vendor estimates). Dupple (2026). The 8 best AI visual inspection tools.
- Bergmann, P. et al. (2019). MVTec AD: a comprehensive real-world dataset for unsupervised anomaly detection. CVPR.
- Roth, K. et al. (2022). Towards total recall in industrial anomaly detection (PatchCore). CVPR. Akiba, T. et al. (2019). Optuna. KDD.
- Bonfiglioli, L. et al. (2022). The Eyecandies dataset for unsupervised multimodal anomaly detection. ACCV. PS-AD (IEEE Access, 2022). Wu, L. et al. (2025). MSDD. Scientific Data 12.
- Photometric-stereo-based defect detection system for metal parts (PSBDDS), 2022. PubMed Central PMC9655976.
- DT Next (Sept 2026). ZEISS picks IIT Madras Research Park as first innovation hub outside Germany.